DESCRIPTION OF COURSEWORK

Course Code	EEE 303
Course Name	Electrical Machinery & Drives
Lecturer	Dr Ng Zi Neng
Academic Session	2023/09
Assessment Title	Laboratory Report

A. Introduction/ Situation/ Background Information

This coursework aims to let students conduct laboratory experiments with electrical machines and analyse their characteristics. Students are required to write a report after each lab activity.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 2 Investigate the expected performance of DC machines, synchronous machines and induction motor.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

Students are required to form a group consisting of 4 members. You will deal with high voltages in this laboratory activity, do not make or modify any connections with the power on. Complete all lab activities as quickly as possible to avoid the rise of machine temperature under load condition. The machines are heavy, practice safety precautions when reallocating them. It is utmost important to work in a group and every member has a role in making sure that the lab activity goes smoothly. Always ask the lab staff to check the connection before turning on the main power.

Lab venue: A1-214 Electrical Power Laboratory

Station: Electrical Machine System



Figure 1. Electrical Machine System

Students are required to perform the following lab activities and submit the reports in pdf format (one report per lab activity) in Moodle by the given due date of submission. This is a group work and only one report submission is required per group for each lab activity.

Proper formatting and style are expected for the lab report:

- Use the Laboratory Report Cover Page and fill in each group member's name and other details.
- Contains following sections: Introduction, Methodology, Results and Discussion, Conclusion, and References.
- Use Times New Roman font with size 12 and 1.5 line spacing.
- Figures and tables have to be captioned and placed according to the standard format, i.e., Figure 1.1 "captions" is placed after the figure and Table 1.1 "captions" is placed before the table.
- Cite references with APA citation format.
- Naming convention for pdf file: Group1_Lab1.pdf

No	Lab Activity	Report submission due date
1	DC Series Wound Machine (Week 3-4)	One week after your lab activity as per selected timeslot, i.e., your group's selected slot for Lab 1 is 11 Oct, Tuesday 2-5pm, submission of this report will be due on 18 Oct, 5pm.
3	Three-phase Rotor Winding Induction Motor (Week 7-8)	
4	Three-phase Salient Pole Synchronous Generator (Week 9-10)	

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	DC Series Wound Machine	6
2.	Three-phase Rotor Winding Induction Motor	7
3.	Three-phase Salient Pole Synchronous Generator	7
	TOTAL COURSEWORK MARK	**20**

F. Task(s)

Refer to Lab Activity Manuals 1-3.

APPENDIX 1

MARKING RUBRICS

Component Title	Lab report				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Formatting, quality of figures and plots	All sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and mentioned in the main text. All figures are clear with important labels and comments. Title and the axis of all plots are well labelled, clear, and easy to read.	Most sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and most are mentioned in the main text. Most figures are clear with labels and comments. Title and the axis of most plots are well labelled and clear.	Some sections are inconsistent in terms of format and style. Most captions for tables and figures are given and mentioned in the main text. Some figures and components in the circuit diagram are not clear and missing important labels and comments. Some title and axis of plots are not well labelled and clear.	Most sections are inconsistent in terms of format and style. Incomplete captions for tables and figures and most are not mentioned in the main text. Most figures and components in the circuit diagram are not clear and missing important labels and comments. Most title and axis of plots are not well labelled and clear.	Messy report and difficult to read with inconsistent format and style. No captions for tables and figures provided. Blurry figures and plots with missing labels provided. Figures and plots are difficult to read and understand.	15	
Introduction and Methodology	Excellently written, concise introduction and methodology with important information that are relevant to the topic.	Well written, concise introduction and methodology with important information that are relevant to the topic.	Averagely written, concise introduction and methodology, consists some irrelevant information.	Averagely written, lengthy introduction and methodology, consists many irrelevant information.	Poorly written, lengthy introduction and methodology, consists many irrelevant information.	15	
Results and discussions	Completed all experiments and data collection. All measured data, calculations, and graphs/plots are correct. Able to evaluate the results with prudent judgment. Analysis is done with excellent description of the relationship involved and explanation with relevant theories.	Completed all experiments and data collection. Most measured data, calculations, and graphs/plots are correct. Results are compared and analysed with good description of the relationship involved and with relevant theories.	Completed most experiments and data collection. Some measured data, calculations, and graphs/plots are incorrect. Results analysis are done with adequate description of the relationship involved.	Completed most experiments, but most measured data, calculations, and graphs/plots are incorrect. Only restatement of the results without commenting on the expected key points.	Experiments not completed. Most measured data, calculations, and graphs/plots are incorrect. Discussion shows little understanding of what the experiment is all about.	50	
Conclusion	The aim of the experiment has been achieved, the key features of the methods used and most important results are summarized.	The aim of the experiment has been achieved, most key features of the methods used and results are summarized.	A conclusion is drawn, but only supported by some experimental evidence.	A conclusion is drawn, but not supported by any experimental evidence.	No sensible conclusion. No comparisons and evidence.	20	
					TOTAL	**100**	

Note to students: Please attach this appendix together with the submission of coursework.